# Netflix Movies & TV Shows

Data Cleaning & Exploratory Data Analysis Report

AnalystLab Africa Data Analytics Internship | Week 2 | Prepared by Zainab Danjuma

## 1. Executive Summary

This project cleaned and explored a dataset of 8,807 Netflix titles (Movies and TV Shows) using Python, Pandas, NumPy, Matplotlib and Seaborn. After cleaning, 8,790 reliable records remained. The analysis shows a movie-heavy catalogue, a content boom between 2017 and 2020, a catalogue dominated by the United States and India, and a strong skew toward mature and teen-rated content.

| | | | |
|---|---|---|---|
| **8,790**<br>clean records | **69.7%**<br>Movies | **2019**<br>peak year (2,016 titles added) | **61%**<br>rated TV-MA or TV-14 |

## 2. The Business Problem

Raw streaming catalogue data is rarely ready for decisions. The Netflix file contained missing directors, casts and countries, text dates and mixed-format fields. Without cleaning and exploration, content strategy questions cannot be answered reliably: What mix of Movies and TV Shows is on offer? How fast has the library grown? Which countries and genres lead? Who is the content built for?

## 3. Dataset Overview

| Item | Detail |
|---|---|
| Source file | netflix_titles.csv |
| Raw size | 8,807 rows x 12 columns |
| Columns | show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, descript |
| Primary key | show_id (system-generated; title is also unique but descriptive and less stable) |
| Numeric / categorical | 1 numeric (release_year); 11 text columns |
| Final size | 8,790 rows x 12 columns, saved as Netflix_Cleaned.csv |

# 4. Data Cleaning

## 4.1 Missing values found

| Column | Missing | % | Action |
|---|---|---|---|
| director | 2,634 | 29.91 | Filled with 'Unknown' |
| country | 831 | 9.44 | Filled with 'Unknown' |
| cast | 825 | 9.37 | Filled with 'Unknown' |
| date_added | 10 | 0.11 | Rows removed |
| rating | 4 | 0.05 | Rows removed |
| duration | 3 | 0.03 | Rows removed |

Director, cast and country are descriptive text fields that cannot be reliably inferred, so they were labelled 'Unknown' to keep the records. Date added, rating and duration are essential for time-based analysis and content classification, and only 17 rows were affected, so those rows were dropped (8,807 to 8,790).

## 4.2 Cleaning steps in order

- **Duplicates:** checked with duplicated(); none found, so no rows were removed.
- **Column names:** stripped spaces, converted to lowercase and replaced spaces with underscores.
- **Date format:** date_added converted from text to datetime64 with pd.to_datetime() after stripping whitespace; zero invalid dates resulted.
- **Data types:** verified; date_added is datetime, release_year is integer, all others are text.
- **Validation:** type (Movie and TV Show only), ratings (14 categories), duration (minutes for movies, seasons for TV shows) and release years (1925 to 2021) all checked and found consistent.

## 4.3 Cleaning summary

| Issue | Action taken |
|---|---|
| Missing values | 'Unknown' for director, cast, country; rows dropped for date_added, rating, duration |
| Duplicates | None found; no rows removed |
| Column names | Lowercase, underscores, no extra spaces |
| Date format | date_added converted to datetime |
| Validation | Type, rating, duration and release year checked; no invalid records |

# 5. Exploratory Data Analysis

## 5.1 Summary statistics

Release years run from 1925 to 2021 with a mean of about 2014, a median of 2017 and a standard deviation of 8.83 years. More than half of the catalogue was released in 2017 or later. Titles were added to Netflix between 2008 and September 2021.

## 5.2 Movies vs TV Shows

Movies account for 6,126 titles (about 70%) and TV Shows for 2,664 (about 30%).

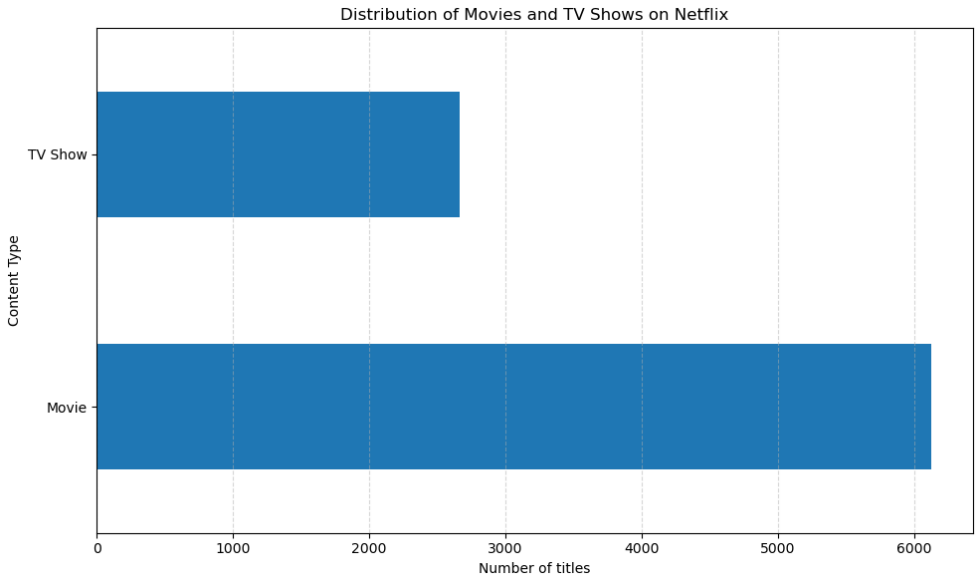



Figure 1: Distribution of Movies and TV Shows

## 5.3 Content added by year

Additions grew from 82 titles in 2015 to 2,016 in 2019, the peak. Volumes then eased to 1,879 in 2020 and 1,498 in 2021. The 2021 figure covers only the months up to late September, so it is a partial year and not a true drop.

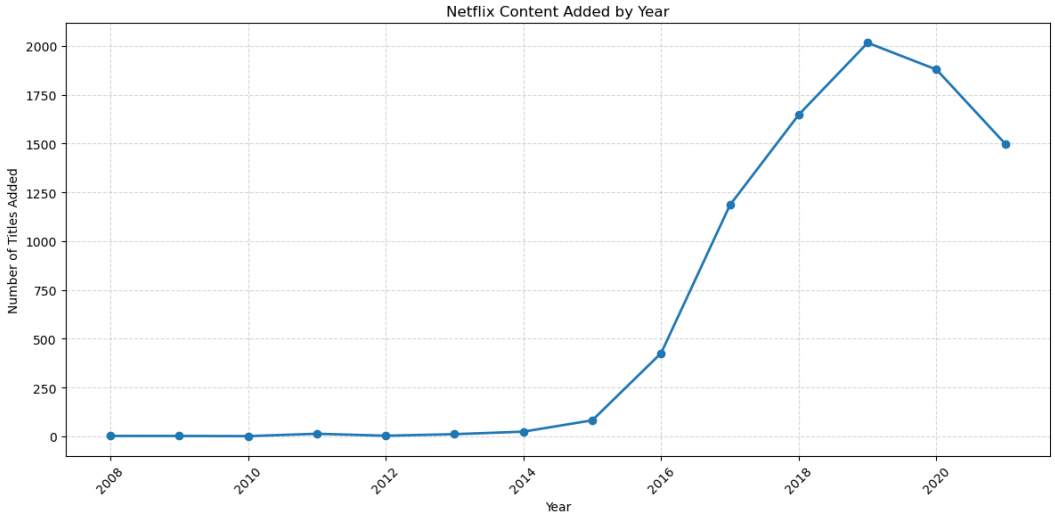



Figure 2: Netflix content added by year

## 5.4 Top content-producing countries

The United States leads with 3,680 titles, followed by India (1,046) and the United Kingdom (803). Canada, France, Japan, Spain, South Korea, Germany and Mexico complete the top 10. Titles with several countries are counted once per country, and titles with unknown country were excluded.

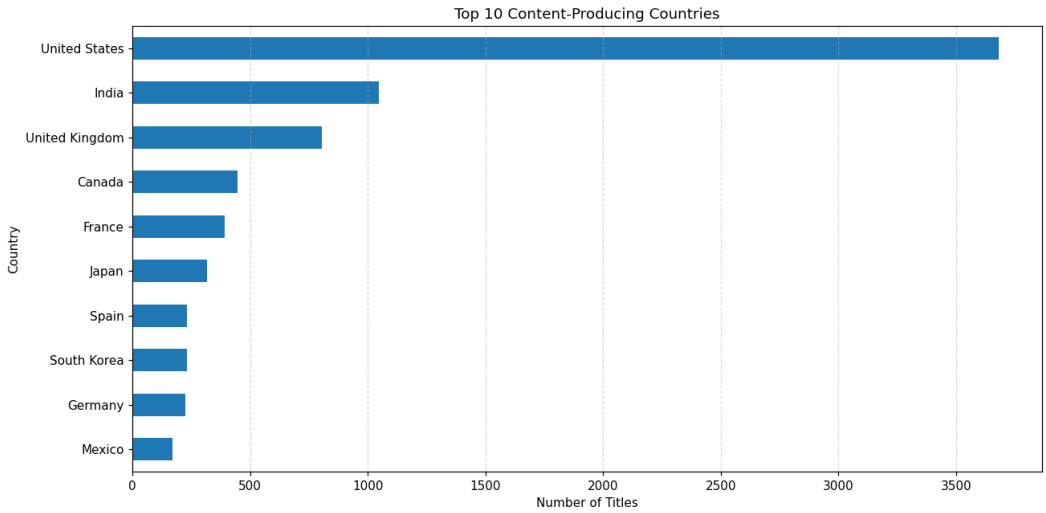



Figure 3: Top 10 content-producing countries

## 5.5 Audience ratings

TV-MA is the most common rating (3,205 titles, 36.5%), followed by TV-14 (2,157). Together they make up about 61% of the catalogue. Content rated for young children (TV-Y, TV-Y7, TV-Y7-FV, TV-G, G) totals 906 titles, about 10%.

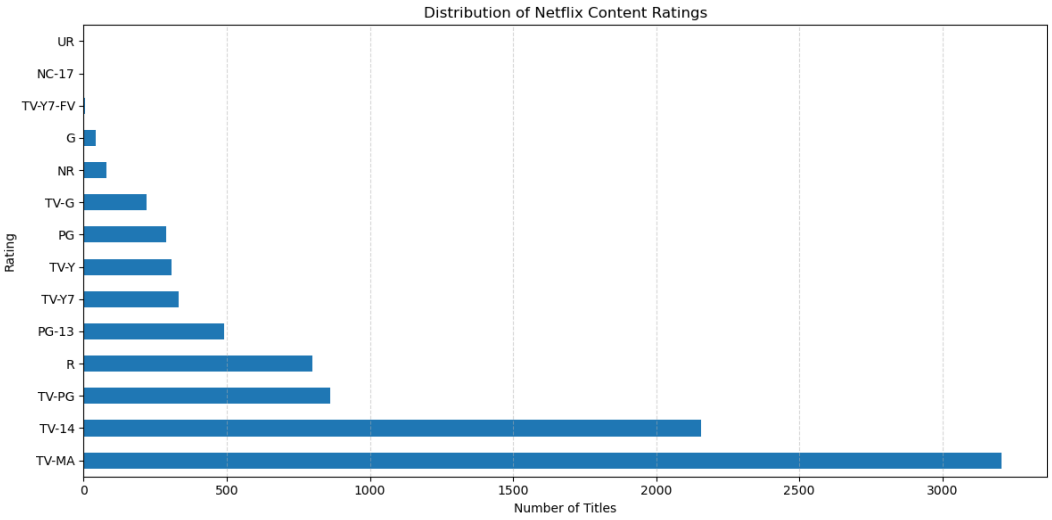



Figure 4: Distribution of content ratings

## 5.6 Top genres

International Movies (2,752), Dramas (2,426) and Comedies (1,674) are the most frequent genre tags. A title can carry several tags, so counts overlap.

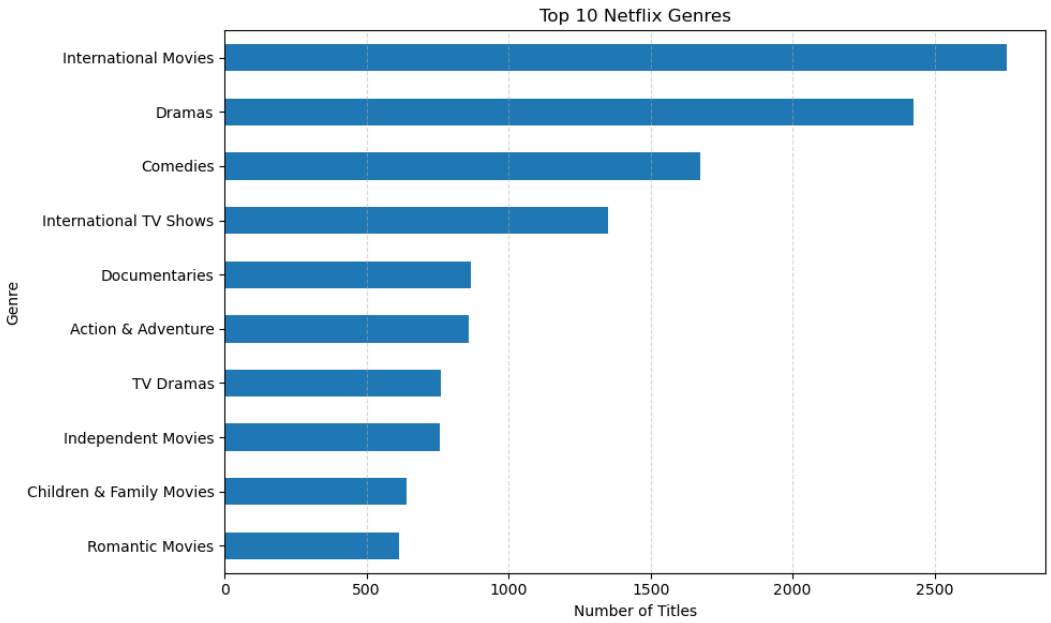



Figure 5: Top 10 genres

## 6. Key Insights

- Movies make up about 70% of the catalogue because Netflix built breadth through films, while TV Shows (30%) are the smaller but more binge-driven part of the library.
- Content additions rose from 82 titles in 2015 to a peak of 2,016 in 2019, reflecting Netflix's rapid global expansion and heavy investment in its library from 2016 onward.
- The United States (3,680 titles), India (1,046) and the United Kingdom (803) lead production, so the catalogue relies on a few large markets while still spanning many countries.
- TV-MA and TV-14 cover about 61% of titles because the platform is built mainly for adult and teen viewers, leaving content for young children at about 10%.
- International Movies and Dramas are the top genres, which shows that global storytelling and drama are the core of the offering.

## 7. Recommendations

- **Keep investing in international content.** It is already the largest genre and supports global growth.
- **Grow underrepresented regions.** Titles from outside the top three markets are a small share, so there is room to diversify.
- **Expand family and kids programming.** Only about 10% of titles target young children, which leaves a gap for households.
- **Protect strength in the US, India and UK** while testing emerging markets such as South Korea, Spain and Mexico.
- **Strengthen TV Show offerings.** At 30% of the catalogue, series are a clear opportunity for retention.
- **Improve data quality at source.** About 30% of titles lack a director and 9% lack a country, which limits deeper analysis.

## 8. Limitations

- The dataset is a catalogue snapshot; it has no viewership, ratings scores or revenue, so popularity cannot be measured.
- 2021 is a partial year (data ends 25 September 2021).
- Multi-country and multi-genre titles are counted once per tag, so totals exceed the number of titles.

## 9. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook.